

CSE 1115: Object Oriented Programming (Spring'25)
Class Test-1 Sec: A Time: 25 minutes Total: 20 marks

Name:	ID:
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1. Write the output of following program:

[10]

```
class Rectangle{
    int length, width;
    static int area = 50;
    Rectangle(int length,int width){
        System.out.println("Length:"+this.length+",Width:"+this.width);
        this.length = length;
        this.width = width;
        System.out.println("Inside the Rectangle Constructor!");
        System.out.println("Length:"+this.length+",Width:"+this.width);
        area += 2;
    }
    {
        length = 10;
        width = 20;
        area -= 10;
        System.out.println("Hello, Area of the rectangle is:"+ length * width);
    }
    int getArea(int length,int width){
        return this.length * width;
    }
}
class Test{
    public static void main(String[] args) {
        Rectangle rectangle = new Rectangle(12,8);
        System.out.println("Area:"+rectangle.getArea(5,15));
        System.out.println(Rectangle.area);
    }
}
```

Output:

Hello, Area of the rectangle is 200(10*20) ->(Initialization Block)

Length:10,Width:20 (first line in constructor) -> values initialized in initialization block

Inside the Rectangle Constructor -> constructor

Length:12,Width:8(Values updated with passed parameters)

Area: 180 (12*15) (this.length=12, but width value is 15 which is passed during function call) -> **focus on this keyword**

42 (initially, area=50, 50-10=40(initialization block), 40+2=42(in constructor))

2. Define a class "Book". Declare 4 member variables

[2]

- i. title(public:String)
- ii. author(public:String)
- iii. price(private:double)
- iv. isbn_no(public:int)

- a. Write a constructor to initialize all the member variables. [2]
- b. Write a display() method which will print all member variables of the class. [2]
- c. Write necessary methods for accessing the private variable. [2]
- d. In main method of "test" class, create an object of vehicle (title="The Tempest", author=William Shakespeare, price=2500,isbn_no=1023234). Now call the display method such that the following output is produced. [2]

Output:

Title: The Tempest

Author: William Shakespeare

Price: 2500\$

ISBN: 1023234

```
class Book{
    public String title;
    public String author;
    private double price;
    public int isbn_no;
    public Book(String title, String author, double price, int isbn_no){
        this.title = title;
        this.author = author;
        this.price = price;
        this.isbn_no = isbn_no;
    }
    public void setPrice(double price){this.price = price;}
    public double getPrice(){return price;}
    public void display(){
        sout("Title: "+title);
        sout("Author: "+author);
        sout("Price: "+price+"$");
        sout("ISBN: "+isbn_no);
    }
}

public class Test{
    psvm(String[] args){
        Book book = new Book("The Tempest","William Shakespeare",2500,1023234);
        book.display();
    }
}
```